

MCODE: Detecting Molecular Complexes in Protein Interaction Networks

Applied to the Human MAP Kinase Interactome (Bandyopadhyay et al., 2010)

CHEM 274B Micro Presentation

Carlos Lopez, Jade Warren, Manula Dombagahawatta, Brendan Dang

What is MCODE (Algorithm Summary)

MCODE: “Molecular Complex Detection”

Uses local neighborhood density to isolate dense regions in a few steps:

- 1) Weigh vertices based on their local network density (“core-clustering coefficient”)
- 2) Predict complexes by seeding a complex with the highest weighted vertex and recursively move out
- 3) (Optional) Post-processing
 - a) Filtering
 - b) Complex size adjustment
 - etc.

Why MCODE matters for molecular sciences

Provides another layer of functional annotation and provides a strategy for navigating large complex datasets with the following benefits:

- 1) Allows for easy visualization of extracted dense regions around a protein of interest
- 2) Once vertex weights have been calculated you can specify a seed vertex and explore high density regions
- 3) Although initially vertex weighing is most time consuming, one can perform subsequent molecular complex predication in $O(V)$ and post-processing in an $O(cV^2)$; where c is the number of complexes found in the complex prediction step. This allows for easy evaluation of different parameters.

Algorithm Overview

1. Compute node weights - $O(VEN^3)$, N = vertex size of the average vertex neighborhood, V = # of vertices, E = # of edges
 - a. Based on k-code & density
2. Choose seed node
 - a. Highest weight
3. Grow molecular complex - $O(V)$
 - a. Add neighbors passing threshold
 - b. Stop when no more pass the rule

Code implementation



The Dataset We Used: Human MAPK interactome

Source: [Badyopadhyay et al., *Nature Methods*, 2010](#)

This data set contains thousand of interactions involving human Mitogen-activated protein kinase (MAPK)-related proteins, which play important roles in cell signaling (i.e., program cell death, cellular behavior in stress responses) and are a focus of study in cancer research and drug discovery.

Paper Identification Methodology of Protein Interactions: two-stage yeast two-hybrid (Y2H) screen

Other notes:

- 641 high-confidence interactions
- Paper assembled a human MAPK network (graph) with nodes as proteins and edges as interactions

Advantages and Limitations

Advantages:

- Modularity
 - Multiple user-defined variables, e.g., core clustering coefficient, density
- Aids visualization with specification of vertex neighborhoods
 - Not all data points have to be in clusters
- Complex finding step is only $O(V)$

Limitations:

- Not highly interconnected data difficult for prediction basis (e.g., Gavin benchmark)
- Lacking temporal and spatial information (e.g., a predicted complex does not necessarily involve proteins that are present at the same time/space)
- Complex stoichiometry not supported, though this could be supported with certain kinds of datasets

Results and Takeaways

By applying MCODE to the MAPK interactome dataset, we were able to extract clusters that represent potential protein complexes. These cluster often contained proteins known to participate unrelated biological pathways.

This demonstrates that MCODE is useful tool for analyzing large biological networks and discovering functional modules within them.

Results:

- Extracted complexes from MAPK network
- Identified dense modules relevant to signaling
- Implementation matches MCODE logic

Conclusion:

- MCODE is effective for PPI module detection
- Useful for pathway analysis & discovering function complexes
- Still alive and well today